

ABDALLAH ELBEHEIRY

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I am a backend engineering student, proficient in .NET and Golang committed to learning new technologies and solving real world problems.

Skills

- **Languages & Platforms:** C#, .NET / ASP.NET Core / Entity Framework Core, Go (Golang), Java, JavaScript
- **Backend & Systems:** RESTful API Design, Multithreading & Concurrency, Systems Programming, Memory Management
- **Databases:** PostgreSQL, SQLite, MongoDB
- **Tools & Infrastructure:** Git, GitHub, Docker, Unit Testing, Integration Testing
- **Software Architecture:** SOLID Principles, Design Patterns (Factory, MVVM, MVC, ...), Clean Code, Test-Driven Development (TDD)
- **Graphics & Simulation:** OpenGL, Physics Engine Development

Projects

QadyJudge — ASP.NET Core, Docker, C#, PostgreSQL — github.com/abdallah-elbeheiry/QadyJudge

- Designed a backend API that securely compiles and executes user code in isolated containers.
- Implemented resource limits and sandboxing to prevent abuse.
- Built RESTful endpoints for code submission, execution, and result retrieval.
- Supported multi-language execution (C, C++, Python).
- Applied Test Driven Programming with integration tests to ensure stability and performance of the project.

Notabor Engine — Go (Golang), OpenGL, GLFW — github.com/abdallah-elbeheiry/NotaborEngine

- Designed a deterministic fixed-tick scheduling system for physics, AI, and input processing.
- Implemented a parallel task execution model using worker pools to run engine systems concurrently.
- Built modular subsystems for rendering, collision detection, input handling, and entity management.
- Integrated OpenGL rendering and GLFW window management, supporting multiple windows and rendering contexts.
- Developed a signal-based input system enabling event-driven actions.

RPGTextAdventureWin — .NET MAUI, C#, SQLite — github.com/abdallah-elbeheiry/RPGTextAdventureWin

- Developed a cross-platform text-based RPG application using .NET MAUI with XAML-based UI.
- Implemented data persistence using Entity Framework Core for game state and player progression.
- Structured application using MVVM-style separation between UI and game logic.
- Adapted and extended an existing open-source project with additional gameplay and UI improvements.

SCICU admission system — ReactJS, MongoDB, ExpressJS, JWT — github.com/MahMoud-Nabil1/FacultyAdmissionSystem-Project

- Contributed to a multi-platform university admission system developed by a team of six developers.
- Implemented core RESTful API endpoints with authentication using JWT.
- Developed the admin dashboard for the web frontend using React.
- Maintained code quality through peer code reviews, documentation, and bug reporting.

Onnet — Java, JOGL (OpenGL) — github.com/fscu-student-team/onnet

- Led a team of 5 developers in building a graphics project using Java OpenGL (JOGL).
- Designed core systems for collision detection, deterministic simulation logic, and memory management.
- Applied software architecture patterns including Builder and Facade to structure rendering and engine components.
- Managed development workflows through GitHub project coordination, pull requests, and code reviews across 330+ commits.

Education

Cairo University, Faculty of Science — Giza, Egypt

B.S. in Computer Science

Expected Graduation: 2027

Relevant Coursework: Operating Systems, Computer Graphics, System analysis, Distributed systems, Databases